EmotiStage: Designing an Emotion-Driven Performance System with Multi-modal Generative AI to Support Emotional Awareness

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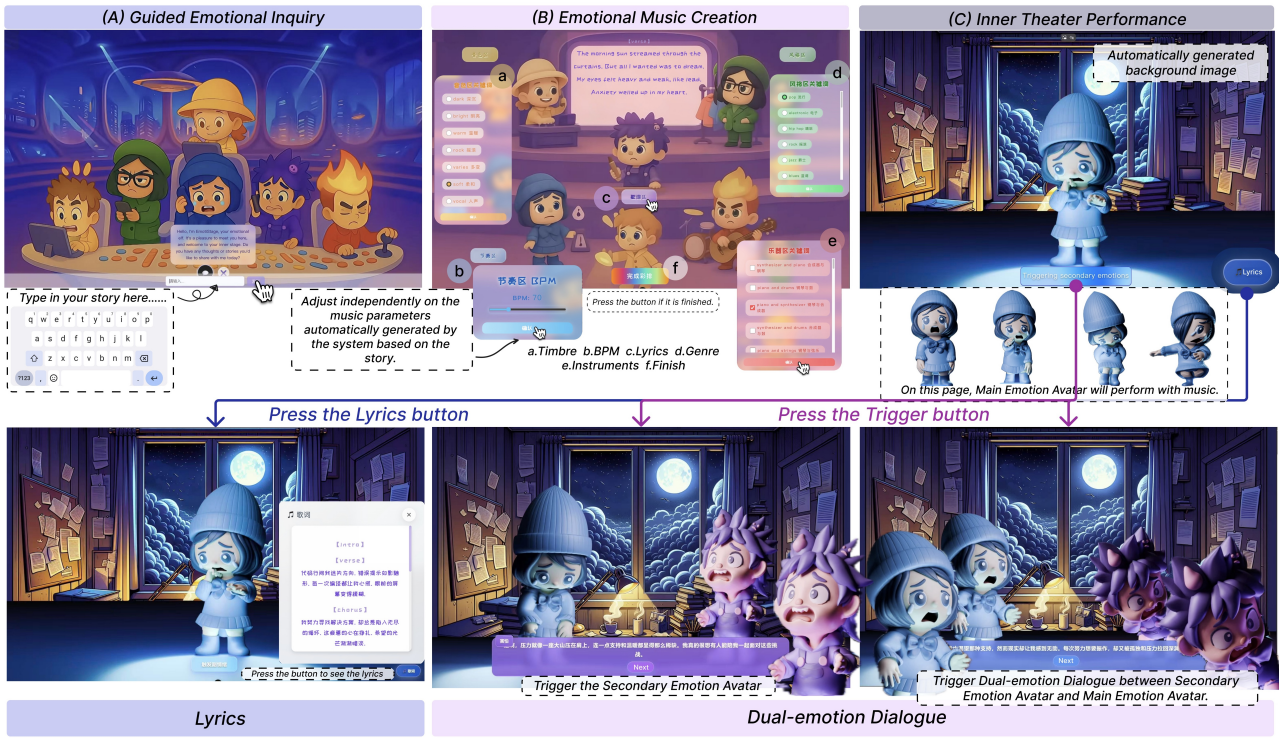


Figure 1: *EmotiStage* is a multi-modal system designed for supporting emotional awareness. The image illustrates the concept of the system: users view a theatrical stage situated inside the brain, where six Emotion Avatars perform as inner theater actors. The stage curtain is textured with brain-like patterns, symbolizing the internal stage of emotions.

Abstract

Emotional awareness is a foundational component of mental well-being, yet existing digital tools often reduce emotional experience to labels, metrics, or inferred states. Such approaches overlook how emotional awareness is constructed through embodied, multi-modal experience and reflective sense-making. We present *EmotiStage*, an emotion-driven performance system powered by multi-modal generative AI that supports emotional awareness by enabling

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users to externalize, observe, and reinterpret emotions through music and stage performance. By guiding users through a creator-to-spectator transition, *EmotiStage* introduces aesthetic and psychological distance, allowing users to re-encounter their internal states as a staged performance. A mixed-method study ($N = 13$) suggests that *EmotiStage* supports emotional awareness by facilitating a progression from expression to observation, helping users visualize and re-experience emotional experiences. We contribute *EmotiStage*, a multi-modal generative AI system for emotional awareness, alongside exploratory empirical findings and design implications for using multi-modal generative arts to support emotional awareness in mental well-being contexts.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; • **Applied computing** → *Consumer health*; **Sound and music computing**; *Performing arts.*

Keywords

Multi-modal systems, Generative AI, Affective computing, Inner theater, Emotional awareness

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1 Introduction

In everyday life, people often sense that something feels off emotionally, yet struggle to trace where it comes from or name what it is. The ability to identify, differentiate, and articulate one's affective states, known as emotional awareness (EA), is foundational to mental well-being [33]. Truly understanding one's emotions goes beyond assigning a single label: it involves perceiving how emotions intertwine, conflict, and are shaped by lived experience [33, 34]. Research on mixed emotions further suggests that people routinely experience simultaneous and even conflicting emotions, particularly during meaningful or self-relevant events, and these co-occurring emotions enrich rather than displace one another [35]. This requires lifting emotions out of the abstract interior and turning them into something that can be observed.

Multi-modal presentation offers a path toward this goal: emotions are themselves composites of multiple sensory signals, and embodied modalities such as music, bodily movement, and visual imagery can effectively restore the valence and texture of emotional experience [13, 14, 17, 30, 48]. Yet existing multi-modal emotion systems fall short of this potential in two fundamental ways. First, modality channels are typically layered independently rather than fused into a unified emotional whole [1, 45]. Second, users are positioned as full creators, required to construct expressive artifacts themselves, and absorbed in the labor of building representations at the cost of the spectator distance that awareness requires [56, 57]. Theater offers a natural resolution: language, music, movement, and scene together form an integrated emotional whole that users can observe from the outside [25, 50]. Yet transforming personal narratives into theatrical forms has traditionally required artistic expertise and production resources beyond everyday reach; multi-modal generative AI now has the potential to lower these barriers.

We present *EmotiStage*, a system that transforms personal emotional narratives into full multi-modal theater performances to support everyday emotional awareness. Users assume the role of *Creator*, first articulating their experience through guided conversation, then iteratively refining an AI-generated musical composition that encodes the emotional tone of the narrative. The system then renders this narrative as a staged performance: Emotion Avatars perform choreographed movement and vocal expression to the generated music against an AI-generated scenic backdrop, embodying the dominant emotional state. Drawing on the understanding that co-occurring emotions are a defining feature of mixed emotions [35], users may subsequently introduce a second Emotion Avatar, whose dramatic encounter with the first externalizes coexisting emotional states as an observable confrontation. The user, now as *Spectator*, observes this performance from the outside.

We evaluated *EmotiStage* through an exploratory mixed-methods user study (N=13). Preliminary quantitative measures showed improvements on the awareness and clarity subscales of the S-DERS and an increase in freely listed emotion words, suggesting the system's potential for cultivating emotional awareness. Participants additionally reported high usability (SUS: 81.35) and perceived empathy (PETS: 78.37). Qualitative findings further revealed how the Creator-to-Spectator structured transition, through the interplay of text, music, and avatar performance, supported participants in identifying, externalizing, and reinterpreting their emotional states.

This work contributes: (1) *EmotiStage*, a multi-modal generative AI system that transforms personal emotional narratives into theater performances to support emotional awareness; (2) exploratory empirical findings on how the Creator-to-Spectator structured transition supports emotional awareness through the interplay of text, music, and avatar performance; and (3) design implications for multi-modal generative AI systems in emotional well-being.

2 Related Work

2.1 Digital Support for Emotional Awareness

Designing digital tools for emotional awareness requires first understanding what higher-level emotional awareness entails. Drawing on Lane and Schwartz's influential theory [33, 34], the most sophisticated levels go beyond naming a single emotion; they involve perceiving *blends*: feeling sad and relieved at once, or sensing the tension between conflicting affective states. This capacity to differentiate and hold multiple co-occurring emotions represents the structural target of emotional awareness cultivation.

Text-based digital tools approach this goal through *active verbal construction*. From chatbot-mediated journaling to LLM-based systems such as DiaryMate [28] and ExploreSelf [51], these tools guide users to articulate emotional experience through language, building a richer vocabulary of feeling [37, 38, 46]. Yet language, by its sequential nature, tends to surface one emotion at a time, and describing a blend requires concepts users may not yet possess. These systems deepen verbal engagement but do not structurally

alter how emotion is perceived: the user remains narrator rather than observer of their own affective complexity.

Artistic modalities offer a structurally different pathway: externalizing emotion into perceivable, multi-sensory form makes the internal landscape observable, allowing coexisting emotions to appear as distinct figures within a shared perceptual space [52, 58]. Systems such as SelVReflect [57] and MoodShaper [55] (VR-based drawing) and the music-based reflection system by Cai et al. [10] demonstrate that such externalization supports awareness in ways verbal reflection cannot. However, constructing an expressive artifact that captures emotional complexity demands creative skill that places this pathway out of reach for everyday users [16, 47].

These two lines of work point to a persistent unmet need: a multimodal artistic container that externalizes emotional complexity in an immediately perceivable form to support emotional awareness, while remaining accessible enough for everyday use.

2.2 Generative Art Media for Emotional Externalization

Research across psychology and the arts shows that different artistic media support emotional externalization through distinct, complementary mechanisms. Music, through physiological entrainment and embodied simulation, translates emotion into something heard and felt rather than merely read, giving it a temporal, rhythmic form [26, 31]. Visual art transforms abstract inner states into symbolic representations that serve as reflective anchors [42, 43]. Drama enables exploration through *dramatic projection* and *aesthetic distance*: projecting the self onto a character lets individuals observe their own affect from a safe third-person perspective [11, 24]. Movement and embodied practices further release implicit emotional tension through the body [20, 29]. These modalities create a coherent multisensory experience that no single medium can achieve.

Recent advances in generative AI have begun to make this integrated pathway accessible: generative models can produce personalized music, imagery, and narrative from minimal input, lowering the skill barriers that once kept artistic creation out of reach [2, 23, 27]. However, existing HCI systems have largely explored single-modality generative outputs, limiting their ability to capture the multi-modal, embodied nature of emotional experience [20, 34]. *EmotiStage* addresses this by integrating conversational input, generative music, and avatar-based theatrical performance into a unified workflow that supports the Creator-to-Spectator transition.

3 EmotiStage

Fig. 2 illustrates the system architecture of *EmotiStage*, which comprises four modules: Guided Emotional Inquiry, Emotional Music Creation, Stage Content Preparation, and Inner Theater Rendering. The details of each module are described in the following subsections. To illustrate the system's interactive components and underlying implementation, we walk through a specific scenario throughout this section: a user experiences stress and internal friction due to a heavy workload and impending deadlines.

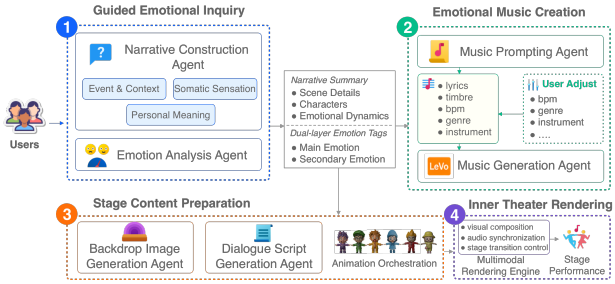


Figure 2: Overview of the *EmotiStage* system architecture. The pipeline integrates multiple LLM-based agents for Guided Emotional Inquiry, Emotional Music Creation, Stage Content Preparation, and Inner Theater Rendering.

3.1 Guided Emotional Inquiry

The user first enters the "Control Room" interface (Fig. 1(A)). The system provides a dialogue window featuring an agent acting as an empathetic facilitator. The user describes feeling anxious and overwhelmed by an accumulation of literature. In the subsequent dialogue turns, the agent guides the user from the objective event (unfinished work) into somatic sensations (tightness in the chest) and finally to personal meaning (doubting one's capabilities). Upon completing the dialogue, the system concludes the inquiry and transitions to the next stage.

System Implementation: As shown in Fig. 2(1), this module is driven by the GPT-4o-2024-11-20 model. The Narrative Construction Agent initiates the interaction with a warm greeting before conducting a structured three- to four-turn inquiry, adaptively extracting the dramatic elements of the narrative: the critical event and its situational context, somatic sensations and bodily manifestations of emotion, and personal meaning and psychological significance. When the user's initial input lacks situational context, the agent first elicits the event background before proceeding through the remaining elements. The agent advances to the next element only after the user responds, enforcing a strict one-question-per-turn constraint to prevent conversational fatigue. The agent employs role-assignment and few-shot exemplars to maintain consistent tone and prevent role drift; all prompt details for this and subsequent agents are provided in Part E of the auxiliary material.

Once the dialogue concludes, the *Emotion Analysis Agent* processes the full conversation history and produces a structured output serving as the shared representation for all downstream generation modules: (1) a *narrative summary* that preserves scene details, characters, and emotional dynamics, and (2) *dual-layer emotion tags* consisting of one Main Emotion (constrained to Sustained Emotions: Happiness, Sadness, Anger) and one Secondary Emotion selected from all six categories. This structured output is serialized and passed to the *Music Prompting Agent*, *Backdrop Image Generation Agent*, and *Dialogue Script Generation Agent* in subsequent stages. In the running scenario, the system outputs "Sadness" as the Main Emotion and "Fear" as the Secondary Emotion.

3.2 Emotional Music Creation

The user then enters the "Rehearsal Studio" interface (Fig. 1(B)). The central screen displays lyrics (Fig. 1(c)) generated based on the

preceding dialogue. The system acts as a musical composer and pre-fills the parameters including Timbre, BPM, Genre, and Instrument (Fig. 1(a,b,d,e)) as default values. The user may adjust any parameter before clicking the "Finish Rehearsal" button (Fig. 1(f)) to initiate the final music generation and stage rendering.

System Implementation: As shown in Fig. 2(2), this module comprises two agents. The *Music Prompting Agent* receives the narrative summary and dual-layer emotion tags produced by the *Emotion Analysis Agent* in the preceding stage, and translates this structured input into two categories of output: (1) *lyrics* grounded in the narrative content and emotional tone of the user's story, and (2) *acoustic parameters* comprising Timbre, Tempo (BPM), Genre, and Instrument. To ensure compatibility with the downstream LeVo model [39], the agent outputs a JSON object conforming to a predefined schema, illustrated by a one-shot exemplar. The backend parses the JSON to pre-fill the control panel with acoustic parameters and display the generated lyrics. User modifications directly update the corresponding JSON fields before the data is passed to the *Music Generation Agent*, which performs end-to-end lyrics-to-melody generation to produce the final vocal track for the performance stage.

3.3 Stage Content Preparation

While the user finalizes the music in the Rehearsal Studio, the system concurrently prepares the stage materials: a scenic backdrop, an avatar dialogue script, and animation assignments, all derived from the narrative summary and emotion tags produced in the preceding stage.

System Implementation: As shown in Fig. 2(3), this module comprises three components.

Backdrop Image. The *Backdrop Image Generation Agent* translates the narrative summary into an image generation prompt, submitted to the DALL-E 3 API, producing a stylized visual scene that encodes the atmosphere of the user's story.

Dialogue Script. The *Dialogue Script Generation Agent* receives the narrative summary and both emotion tags, generating a six-line script following a three-turn dramatic arc of conflict, development, and resolution. The output is a JSON array of six utterances tagged with speaker identifiers, enabling the frontend to render each line sequentially and switch between avatar panels.

Avatar Animation Orchestration. To externalize abstract emotional states, we designed six personified *Emotion Avatars* (Happiness, Sadness, Anger, Surprise, Disgust, and Fear) based on Ekman's basic emotions [15]. Their visual design uses intuitive color-emotion mappings [54] (e.g., red for Anger, blue for Sadness) and simple morphological metaphors to support rapid recognition. We further introduce *Narrative Sustainability*, distinguishing Sustained Emotions (Happiness, Sadness, Anger) that anchor the main tone from Transient Emotions (Surprise, Disgust, Fear) that respond to narrative conflict, enabling structured dual-layer emotional tagging. Avatar selection and animation assignment are rule-based, driven by the dual-layer emotion tags. The Main Emotion label determines the avatar asset and its five-animation repertoire, each pre-rendered as a looping video clip (see Part B of the auxiliary material for selection criteria). Animation timing is driven by the audio player's `timeupdate` event: the total track duration is divided equally across the five clips so that the full animation repertoire completes exactly one cycle over the course of the song.

3.4 Inner Theater Rendering

In the final "Inner Theater," the interface switches to a full-screen stage (Fig. 1(C)). The performance unfolds in two phases. **(1) Emotion Solo Stage:** The Sadness avatar appears first against a dynamically generated backdrop depicting a desk piled with books, performing the generated rock music through singing and physical movements such as sighing and shaking its head, embodying the dominant emotional tone of the user's narrative. **(2) Dual-Emotion Dialogue:** The secondary emotion is activated by the user via a "Trigger Secondary Emotion" button. Upon clicking, the screen splits vertically as the Fear avatar enters from the right. The two avatars then perform a scripted dialogue in which Sadness expresses exhaustion while Fear voices anxiety about upcoming deadlines, externalizing these coexisting emotions as a dramatic on-stage confrontation.

System Implementation: As shown in Fig. 2(4), the Vue.js frontend composites all modalities at runtime: avatar video layers are rendered over the backdrop image via CSS compositing, and audio is managed by the Web Audio API. Each modality addresses a distinct dimension of the emotional story: the backdrop renders the narrative setting; the vocal track carries the emotional tone and lyrical content; the avatar animations embody the dominant emotion through movement; and the dialogue gives structural form to the tension between coexisting emotions. Upon the user's trigger, the frontend transitions to a split-screen layout, loading the secondary avatar into the right panel alongside the continuing main avatar.

4 Evaluation

We conducted a mixed-methods user study with 13 participants to explore how users engaged with *EmotiStage* and to gather preliminary evidence of its potential to support emotional awareness, usability, and perceived empathy. The study was approved by the Institutional Review Board (IRB) at Duke Kunshan University (Protocol #2025ZW103).

4.1 Procedure

The study was conducted online, with each participant using their own laptop in a comfortable, quiet, and private space. All interactions were self-guided to allow independent engagement with the system, while the researcher remained available via Zoom for on-call technical support. This setup minimized interference while ensuring timely assistance.

Before the study, participants provided informed consent and completed a demographic questionnaire and the S-DERS (baseline emotion perception/awareness in the chosen story), and listed identified emotions. They then received three instructional pictures of *EmotiStage*'s functionalities (*Guided Emotional Inquiry, Emotional Music Creation, and Inner Theater*) and familiarized themselves with the system. Participants then engaged with *EmotiStage* in three phases.

Afterward, participants reconnected with the researcher via Zoom, completed the S-DERS again, and freely listed emotions to detect any observable shifts in awareness and clarity. They then

filled out the SUS (System Usability Scale) and PETS (Perceived Empathy of Technology Scale), followed by a semi-structured interview (about 22 minutes) on their experience and perceptions. The total duration per participant was 50 to 60 minutes, covering pre-study, *EmotiStage* engagement, and post-study activities.

4.2 Participants

We recruited 13 participants through social networks and snowball sampling [3, 18]. Our sample consisted of 8 females and 5 males with an age range from 19 to 27 years in China. They were also asked to prepare a personal story they were willing to reflect on during the study, ensuring they could reflect on and articulate emotions embedded in the story. We provided compensation of 100 RMB to all participants who participated in the evaluation.

4.3 Data Collection

For each participant, we collected system interaction data, comprising their input stories and detailed interaction logs. We collected quantitative data using three validated questionnaires and additionally gathered open-ended emotion data by asking participants to freely list the emotions they perceived in the story both before and after using the system. To gain a deeper understanding of participants' experiences, we also conducted semi-structured interviews.

4.3.1 Questionnaires. We employed the State Difficulties in Emotion Regulation Scale (S-DERS) [36] to detect observable shifts in participants' emotional awareness immediately before and after their interaction with *EmotiStage*. Unlike trait-based measures of emotion regulation that focus on stable and dispositional tendencies, the S-DERS is designed to assess fluctuations in emotion dysregulation over brief intervals, making it particularly suited for evaluating short-term changes around a specific emotional event [36]. For our purposes, we focused on two subscales most relevant to emotional awareness: Lack of Awareness of Current Emotions (AWARENESS; Items 3, 10, 16, 22, 26) and Lack of Clarity about Current Emotions (CLARITY; Items 11, 19). The other subscales, Non-acceptance of Current Emotions (NON-ACCEPTANCE) and Limited Ability to Modulate Current Emotional and Behavioral Responses (MODULATE), were excluded as they pertain to emotion regulation strategies rather than awareness and clarity of emotional states. The responses to these questions are based on a seven-point Likert scale.

After using *EmotiStage*, participants completed two post-study questionnaires. The System Usability Scale (SUS) [6] measured overall system usability, while the Perceived Empathy of Technology Scale (PETS) [49] assessed how empathic the system was perceived to be, covering Emotional Responsiveness, Understanding, and Trust.

4.3.2 Free-Listing of Emotions. To quantify observable shifts in emotional awareness, participants freely listed all emotions in their story before and after using *EmotiStage*. This open-ended method captured two facets: clarity (precision of identification) and variety (range of distinct emotions). Free-listing is widely used in HCI as a complementary quantitative measure in emotion-related systems [9, 22, 53]. Comparing pre- and post-lists allowed us to assess whether participants' awareness became more differentiated and nuanced.

4.3.3 Interview Protocol. In addition, we conducted semi-structured interviews averaging 22 minutes in length (*range*: 14:24 to 33:45). Participants were invited to reflect on their previous use of emotion-related applications, share their overall perceptions of *EmotiStage*, and describe how its key features, *Guided Emotional Inquiry, Emotional Music Creation, and Inner Theater Performance*, shaped their emotional awareness. Conversations also explored the interplay of these features, potential scenarios for future use, and any shifts in participants' understanding of their emotions. The complete interview script is provided in Part A of the auxiliary material.

4.4 Data Analysis

4.4.1 Quantitative Analysis. Given the exploratory nature of the study and the small sample size, quantitative results are interpreted as preliminary indicators of the system's potential rather than definitive evidence of efficacy. Data normality was first verified using the Shapiro-Wilk test [19]. To evaluate observable shifts in emotional awareness and granularity, we compared pre- and post-experience scores for both the S-DERS and the free-listing task (number of distinct emotion words) using paired-samples t-tests [21]. For all significant comparisons, we reported mean differences (M_{diff}) and effect sizes (Cohen's d) to quantify the magnitude of the intervention's impact. For SUS, we followed Brooke's scoring procedure [5, 6, 8], converting item responses, summing, and multiplying by 2.5 to obtain a score from 0 to 100. For PETS, we used the validated 10-item two-factor structure (PETS-ER: Emotional Responsiveness; PETS-UT: Understanding and Trust) with a 101-point slider, and computed subscale scores following the validated method [49].

4.4.2 Qualitative Analysis. All audio recordings from the semi-structured interviews were transcribed verbatim. We conducted a thematic analysis of the semi-structured interview data following Braun and Clarke's reflexive thematic analysis framework [7]. Two researchers independently conducted initial coding, followed by team-based review and refinement. This iterative process continued until consensus on final themes was reached and data saturation was achieved, with no new codes or distinct insights emerging [7, 41]. In parallel, we employed automated Chinese word segmentation to categorize narrative inputs into five thematic categories: Relationships, Leisure and Positive Moments, Self-development Challenges, Health, and Daily Hassles, with results reviewed by the research team to ensure semantic validity.

5 Result

5.1 Usability and Perceived Empathy

EmotiStage demonstrated good usability, achieving a mean SUS score of 81.35 (SD = 13.05), corresponding to a "Good" rating [5]. Eleven of 13 participants rated the system as easy to use (median = 5 on a 5-point Likert scale), suggesting the interaction did not impose excessive cognitive load. As P5 noted, *"The system allowed me to complete the inner theater creation through simple and intuitive interactions, which turned out to be much more convenient than I expected."*

Perceived empathy scores (PETS) averaged 78.37 (SD = 15.31), with subscale scores of 80.76 (SD = 13.68) for Emotional Responsiveness and 74.79 (SD = 20.04) for Understanding and Trust, indicating

that participants generally perceived the system as empathic across both dimensions.

Participants engaged with narratives spanning diverse everyday contexts, with *Relationships* (46%), *Leisure and Positive Moments* (38%), and *Self-development Challenges* (31%) as the most common themes, extending also to *Health* (23%) and *Daily Hassles* (15%), demonstrating the system's versatility across emotional domains (see Part C of the auxiliary material for details). At the level of avatar casting, Sadness appeared five times and Happiness and Anger each appeared four times as the Main Avatar, indicating that participants used the system for diverse motivations spanning both positive and negative emotional experiences. At the level of Secondary Avatar selections, participants also surfaced emotions entirely absent from the Main Avatar category: Surprise was the most frequent (4 cases), followed by Disgust (2 cases) and Fear (1 case). These Secondary Avatars often co-occurred with contrasting main emotions (e.g., disgust embedded within an anger narrative), reflecting the blended emotional structures that *EmotiStage* was designed to surface.

5.2 Facilitating Emotional Awareness

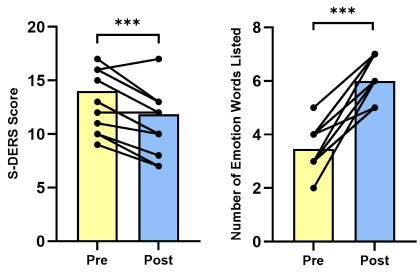


Figure 3: Participants' emotional awareness before (Pre) and after (Post) using *EmotiStage* (* p < .001). Left: S-DERS awareness/clarity scores. Right: number of distinct emotion words listed. Bars show mean ± SEM; lines show paired data.**

5.2.1 Overall Emotional Awareness. Paired t-tests on the S-DERS Awareness and Clarity subscales showed significantly improved scores after system use ($t(12) = 5.53$, $p < .001$, $M_{\text{diff}} = 2.15$, $d = 1.53$), suggesting that the system shows potential to assist users in establishing recognition and clarity of their emotional states within a short-term interaction. A significant increase in freely listed distinct emotion words was also observed ($t(12) = 10.44$, $p < .001$, $M_{\text{diff}} = 2.54$, $d = 2.90$), suggesting that *EmotiStage* may support growth in emotional granularity (Fig. 3). Qualitatively, P1 described the experience as opening "*new ways to think about emotions I had not considered before.*" To uncover the mechanisms underlying these preliminary findings, we present a qualitative analysis of participant interviews across the three core modules.

5.2.2 Guided Emotional Inquiry: Supporting Initial Emotional Articulation. Participants consistently described the *Guided Emotional Inquiry* as a structured support that helped organize fragmented emotional knowledge through two complementary mechanisms.

First, participants reported that the sequential prompts functioned as retrieval cues, surfacing overlooked narrative fragments. Several participants noted that they initially focused on a single dominant event, overlooking concurrent sub-events with contrasting emotional valence. P4, who initially described only frustration at a coding bottleneck, reported being prompted to recall concurrent user praise and reaching 20k followers: "*I had completely filtered out the good part because of the stress… The question helped me realize it was not just frustration, but a mix of pain and joy.*" Similarly, P5 reported that the inquiry prompted him to recall his relative's "active mindset" during a family visit, allowing him to recognize an underlying sense of "delight" that had been obscured by his initial label of "surprise."

Second, participants reported that targeted questions helped ground diffuse thoughts in concrete situational attributions. P12 initially framed her experience in global terms ("*I am an unlucky person…*"). After the system anchored on a specific event and prompted "What happened after that?", she reported reframing her reflection toward situational meaning: "*I realized my pain came from feeling violated and disgusted by that experience, not because there is something wrong with me.*"

5.2.3 Emotional Music Creation: Channeling Emotions into Musical Self-Reflection. Interaction logs indicate that 12 of 13 participants actively modified the AI-generated defaults, with BPM as the most frequently adjusted parameter (10/12) (see Part D of the auxiliary material for details). Participants reported adjusting *Genre* and *Timbre* to align with their emotional identities, such as selecting rap to express anger (P13) or a brighter timbre to convey joy (P8). As P9 noted, "*Without this fine-tuning step, the generated music would feel slightly off from my inner thoughts.*"

Beyond personalization, participants reported framing the adjustment process as a form of emotional self-verification. Iterating through parameter combinations prompted reflection on whether the sonic output accurately matched their internal state. P2 reported: "*I thought, 'Is this warm timbre really my emotion?' It sounded incorrect, so I switched to a colder one.*" P7 described the process as "*repeatedly questioning myself, helping me confirm what kind of emotion I was actually in.*" These accounts suggest that AI-generated defaults served as a provisional starting point for iterative emotional sense-making.

5.2.4 Inner Theater: Externalizing and Deepening Emotional Awareness. Upon entering the *Emotion Solo* stage, participants (P1, P3, P5, P8, P12, P13) reported a perceptual shift from first-person involvement to a third-person view, redirecting attention from recounting events to perceiving emotional intensity as an external object. P7 observed that the system "*dissolved the story into the expression of emotion*," while P6 described the *Emotion Avatars* as making abstract feelings "*vividly stand in front of me.*"

Notably, the avatars' exaggerated movements reconnected cognitive descriptions with latent bodily sensations. Although P3 had described her bodily reaction as "*nothing special*" during the inquiry, witnessing the anger avatar "*roaring and swinging its fists*" prompted her to recall that she had actually been trembling with anger, a physical sensation she had forgotten until seeing it enacted. Participants also reported that music and lyrics functioned as an aesthetic container that made intense emotions more approachable, with P10 noting that "*when it became lyrics, it felt acceptable.*"

The *Dual-Emotion Dialogue* extended this process by surfacing latent emotions that participants had not initially recognized. The

introduction of a *Secondary Emotion Avatar* disrupted participants' initial emotional framings and invited reflection beyond a single emotional perspective. P8 and P9 reported that their initial emotional labels felt incomplete after the secondary avatar expressed surprise, recognizing they had "*overlooked transient emotions.*" P12 similarly discovered an unrecognized anger: "*Before, I thought it was just awkwardness or helplessness, but when the secondary avatar questioned, 'This action completely disregarded my feelings!', I realized anger was lurking within my chaotic emotions.*"

Participants further reported reflecting on the relationships between coexisting emotions in varied ways. Some (P4, P10, P11) perceived the dialogue as a conflict between opposing forces, while others (P3, P5) described the two avatars as complementary perspectives on the same experience. Several participants also situated emotions temporally, with P1 reflecting: "*Sadness might be what I feel now, but at that moment, anger was much stronger.*" These accounts suggest that participants moved beyond labeling discrete emotions toward perceiving how distinct emotional states coexist, conflict, and shift across time within a single experience.

Finally, participants reported that the stage backdrop reinforced emotional understanding by externalizing narrative context. P11 noted that environmental details, including the dentist's office, the heartbeat, and the shadowless lamp, were integrated into the backdrop, clarifying what had triggered their tension. These findings suggest that the multi-modal staging of the Inner Theater offered participants an externalized vantage point from which the blended and relational structure of their emotions became perceivable, supporting a shift from surface-level labeling toward a more differentiated understanding of their emotional experience.

6 Discussion

6.1 Two Modes of Emotional Awareness: The Theater as Structural Condition

Our findings suggest that *EmotiStage* supports two qualitatively distinct modes of emotional awareness, and that the theatrical framework is the structural condition that makes this possible.

Theater differs from other artistic media in its fundamental temporality and uncontrollability [12]. A performance unfolds before the audience without the possibility of pause, revision, or return. *EmotiStage* introduces this quality into emotional processing: in the first mode, during the articulation stage, users retain narrative control, organizing and naming their experiences through a form of self-disclosure. This is the process of *saying*, in which awareness is actively constructed through articulation. In the second mode, when AI transforms this input into a complete performance, what users encounter is no longer a version they fully anticipated or authored. In this spectatorship moment, emotional awareness is passively triggered, closer to *recognizing* than to expressing.

This second mode aligns with the psychological mechanism of self-distancing [4, 32], whereby adopting an observer's perspective rather than an immersed first-person stance deepens emotional understanding. Yet what *EmotiStage* produces goes beyond a shift in perspective. The AI-mediated artistic translation generates an estrangement effect, allowing users to perceive what had become invisible through overfamiliarity. Prior emotion-aware systems, whether journaling tools [28, 51] or expressive art platforms [55, 57], position users as the consistent subject and author of their own narrative, sustaining the same underlying logic of awareness throughout. *EmotiStage*'s linear transition from creator to spectator opens a gap in which narrative control is temporarily relinquished, and it is within this gap that users reach what active storytelling cannot access: forgotten emotional fragments, tensions and overlaps between feelings, and affective textures that language has not yet captured. The depth of this recognition suggests that future emotional support systems should consider how to leverage loss of control as a design resource, rather than treating artistic media solely as outlets for emotional expression.

6.2 Design Implications for Multi-modal Generative AI in Mental Well-being

Both modes, *the active construction of awareness through articulation* and *the passive triggering through spectatorship*, are not incidental to *EmotiStage*'s design; they emerge from specific multi-modal design decisions that point to broader implications for affective computing systems.

6.2.1 Form-Grounded Multi-modal Integration as a Design Principle. Participants' feedback revealed how multi-modal integration operated within the Inner Theater: background imagery established the setting and atmosphere of the user's story; lyrics were distilled from the emotional narrative, while the avatar's bodily movements synchronously interpreted them; when a secondary emotion avatar entered, the dialogue conflict between the two avatars externalized the most complex and tension-laden emotional moment as a visible dramatic confrontation. Participants noted that these modalities worked together in shaping their perception of the emotional story, rather than independently conveying separate aspects of emotional information, suggesting that each modality, anchoring a different narrative dimension from the same story, collectively formed a perceivable emotional whole.

Most existing multi-modal affective systems treat modalities as independent information channels, processing them in parallel and integrating them through fusion strategies [1, 44]. While effective for conveying emotional information, this logic is less capable of restoring the inherent wholeness of emotional experience. Prior research suggests that emotion does not pre-exist internally and then get transmitted through separate sensory channels; rather, it emerges through the coordination of sound, movement, language, and perception [20]. When modalities are decomposed and reassembled, this fundamental wholeness is lost.

EmotiStage's design points to an alternative approach, which we term form-grounded multi-modal integration: **rather than asking which modalities to select and how to fuse them, designers should seek a human-meaningful composite form that naturally governs the relationships among modalities**. In practice, this means selecting a form-level framework in which the role of each modality is already defined by the form's internal logic, then mapping system outputs onto those roles. Theater provided this structure in *EmotiStage*; designers working in other affective contexts might identify analogous composite forms, such as ritual or ceremonial narrative, that carry similarly pre-structured relationships among sensory channels. For affective support systems,

this suggests shifting design attention from the technical fusion of modalities toward the selection of a form-level framework that gives those relationships human meaning.

6.2.2 Delegating Artistic Rendering to Preserve Emotional Ownership. Our study suggests that the boundary between what users control and what AI generates is an emotional decision as much as a usability one. In *EmotiStage*, users provide a narrative seed and calibrate affective parameters, while AI handles the technical execution of melody, animation, and scenery. Participants did not experience this delegation as a loss of authorship. They valued the separation: retaining ownership over emotional meaning while releasing responsibility for artistic rendering allowed them to stay oriented toward their inner experience rather than toward the mechanics of production. When users are burdened with low-level creative decisions, attention shifts away from emotion and toward execution, which works against the reflective stance that emotional awareness requires [40].

For multi-modal generative AI systems in emotional well-being contexts, the division of labor between user and AI should be calibrated so that what users control remains expressive and emotionally legible, while what AI generates remains surprising enough to sustain the estrangement effect that deeper awareness depends on. We suggest that **designers identify this boundary in relation to the specific emotional goals of the system, attending to where delegation begins to undermine the sense that the output still belongs to the user**.

6.3 Limitations

This work has several limitations. *First*, the evaluation involved 13 university students in China and a single-session study without a control condition. Consequently, the findings should be interpreted as preliminary evidence of the feasibility and user experience of *EmotiStage*, rather than causal evidence of its effectiveness or its generalizability across populations. Future work should conduct larger randomized longitudinal studies with more diverse participants to examine sustained effects and cultural differences in how theatrical emotional representations are perceived. *Second*, *EmotiStage* focuses on situational emotional awareness by helping users externalize and interpret a single emotional episode. It does not yet support structural emotional awareness, such as identifying recurring emotional patterns across time. Future systems could incorporate affective memory and replay mechanisms that archive Inner Theater performances as longitudinal emotional records, enabling reflection on evolving emotional trajectories. *Third*, the current implementation relies on predefined animation assets, stage effects, and coarse emotion categories based on Ekman's six basic emotions. While these design choices provide a consistent theatrical language, they constrain expressive richness and may not fully capture nuanced or blended emotional experiences, potentially reducing coherence across modalities. Future work could investigate more expressive generative animation, richer emotional representations, and adaptive multi-character narratives that better reflect the complexity of everyday emotions. *Finally*, several generative components, particularly emotion-to-music mapping, remain inherently subjective and only partially interpretable. Rather than treating these mappings as objectively correct, *EmotiStage* encourages users to iteratively adjust generated outputs to better match their internal experiences, making refinement itself part of the reflective process. Future research could further improve transparency and user agency by developing more interpretable and controllable multi-modal generation mechanisms.

7 Safe and Responsible Innovation Statement

EmotiStage operates in a sensitive domain of AI generation, taking private emotional narratives as input and reflecting them as AI-generated performances. We addressed ethical risks through several measures. At the system and prompt level, the Guided Emotional Inquiry agent enforces a one-question-per-turn constraint and maintains a warm, non-judgmental tone, eliciting narrative reflection rather than clinical disclosure. At the recruitment level, participants with current emotional or mental health conditions were screened out, and the study was conducted in private settings with researcher support throughout, with IRB approval.

8 Conclusion

This paper presented *EmotiStage*, a multi-modal generative AI system that transforms personal emotional narratives into theatrical performances through three integrated stages: Guided Emotional Inquiry, Emotional Music Creation, and Inner Theater. Beyond introducing a novel system, this work demonstrates how generative AI can support emotional awareness by translating internal experiences across complementary artistic modalities. The creator-to-spectator transition enables two distinct forms of awareness: reflective awareness emerging through self-expression, and perceptual awareness arising when users observe their emotions reinterpreted as an external performance. By combining dialogue, music, visual imagery, and embodied avatars, EmotiStage provides a multisensory medium for externalizing emotional complexity that extends beyond language alone. An exploratory mixed-methods study with 13 participants provided preliminary evidence of improved emotional awareness and granularity, together with high usability and perceived empathy. As generative AI increasingly becomes a medium for creative expression rather than information delivery, we envision AI systems that not only understand emotions but also transform them into experiences that foster reflection, meaning-making, and psychological growth.

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